Multinetwork Algorithm for Coastal Line Segmentation in Remote Sensing Images

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***Abstract*— The demarcation between the sea and the land, commonly referred to as the coastline, is of paramount importance for the dynamic monitoring of its alterations. This monitoring is essential for the effective utilization of marine resources and the conservation of the ecological environment. Addressing the challenges posed by the extensive expanse of coastal lines, which can complicate their acquisition and processing, this study utilizes remote sensing imagery to introduce an algorithm for coastal line segmentation. The algorithm integrates multiple networks to enhance its effectiveness. Innovations encompass the development of an extraction algorithm for coastal lines that are as follows. First, utilize an attention-guided conditional generative adversarial network (AC-GAN) model, which redefines the task of image segmentation by framing it as a style transformation problem. Second, a strategy for coastal line segmentation utilizes Dense Swin Transformer Unet (DSTUnet) to construct a densely structured model. This approach integrates Transformer to prioritize focal regions, thereby enhancing image and semantic interpretation. Third, a transfer learning framework is proposed to integrate multiple features, leveraging the strengths of different networks to achieve accurate segmentation of coastal lines. The study introduced two datasets, and the experimental results confirm that parallel network configurations and asymmetric weighting are superior in achieving optimal results, with an area overlap measure (AOM) score of 85%, outperforming the Unet by 5%.**

***Index Terms*— Coastal line, generative adversarial network (GAN), remote sensing, segmentation, transformer, Unet.**

Manuscript received 24 February 2024; revised 16 June 2024 and 4 July 2024; accepted 27 July 2024. Date of publication 30 July 2024; date of current version 12 August 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFB3905700, in part by the Western Light of China under Grant XAB2022YN10, and in part by China Postdoctoral Science Foundation under Grant 2023M740760. *(Corresponding authors: Xing Wang; Huping Ye.)*

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Digital Object Identifier 10.1109/TGRS.2024.3435963

I. Introduction

THE coastal line, which delineates the interface between land and sea, includes both continental and island coastlines. Subject to the impacts of tectonic movements, global warming, tidal fluctuations, and human activities, coastal areas are continuously changing. The dynamic monitoring and timely extraction of coastal lines have significant implications for the utilization of marine resources [1] and the conservation of ecological environments [2].

The earliest techniques used for measuring coastal lines predominantly relied on fieldwork, which was known for its extended duration, high labor requirements, and inherent risks associated with the measurement procedures. In recent years, the development and progress of remote sensing technology have enabled researchers to access extensive amounts of data for research purposes. Remote sensing is a technology that enables noncontact and long-distance detection by utilizing sensors to analyze the radioactive and reflective characteristics of electromagnetic waves emitted by objects. Semantic segmentation can realize the segmentation of the concrete meaning of the image [3]. Current algorithms utilized for coastal line analysis through remote sensing technology have been discussed. Sun et al. [4] reviewed the current state of coastal zone analysis using remote sensing, confirming its feasibility and effectiveness. Yasir et al. [5] employed a combination of remote sensing and GIS technology to identify coastal boundaries and monitor alterations in their respective areas. Domazetović et al. [6] conducted a specialized tool for extracting coastal lines to retrieve coastal line data. Paravolidakis et al. [7] utilized edge detection and optimization algorithms to expedite the extraction of coastal lines. Modava and Akbarizadeh [8] carried out spatial fuzzy clustering and the active contour method using SAR imagery to extract coastal lines. Dai et al. [9] conducted coastal zone extraction by analyzing repeat high-resolution satellite imagery. Liu et al. [10] proposed an analysis of coastal lines using Landsat-8 OLI imagery. Alcaras et al. [11] delineated coastal boundaries by analyzing optical imagery. Zhu et al. [12] utilized multispectral techniques by integrating prior edge information with active contour models to extract coastal lines. Zhou et al. [13] conducted a review on coastal line extraction utilizing remote sensing data. Iqbal et al. [14] delineated coastal boundaries by analyzing SAR imagery through Doppler centroid images. Çelik and Gazioğlu [15] introduced machine learning techniques to develop classifiers for the extraction of coastal

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TABLE I
CHARACTERISTICS OF DIFFERENT METHODS FOR COASTLINE SEGMENTATION

Methods	Advantages	Disadvantages
Traditional feature	Models are established based on interpretable and quantifiable features such as color and texture.	Most of methods require manual interaction, and their performance is challenging to further enhance to a certain extent.
Deep feature	By leveraging a large amount of annotated data to bridge the gap between image and semantics, segmentation can be achieved with good results, representing the primary direction of current research.	The similarity between the testing and training data significantly influences the segmentation effectiveness.

lines. Prabhu and Murugan [16] proposed to segmentation of coastline based on satellite remote sensing data. Lu et al. [17] delineated coastal boundaries using high-resolution remote sensing imagery. Jing et al. [18] proposed a joint-learning boundary and segmentation network for coastal line analysis. Yu et al. [19] employed a thresholding method to extract coastal lines. Zollini et al. [20] utilized an active connection matrix (ACM) image enhancement technique for the extraction of coastal lines. Alcaras et al. [21] employed an unsupervised classification approach for extracting coastal lines. Mirsane et al. [22] integrated radar and optical imagery and utilized wavelet features to extract coastal lines. Jin et al. [23] utilized a range of threshold segmentation algorithms for extracting coastal lines. Randazzo et al. [24] employed high-resolution imagery to extract coastal lines. Qiu et al. [25] developed a multifeature fusion model utilizing multiband imagery to extract coastal lines. Modava and Akbarizadeh [26] conducted an analysis of coastal lines utilizing the level set algorithm. Sun et al. [27] utilized Quadtree decomposition to extract coastal lines at various scales. Aghdami-Nia et al. [28] enhanced the Unet model for the extraction of coastal lines. Alcaras et al. [29] employed K-means and K-nearest neighbor algorithms for coastal line extraction. Yang et al. [30] proposed a method for extracting coastal lines under complex conditions. Dervisoglu et al. [31] conducted a comparison between pixel- and object-level algorithms for coastal line segmentation. Bamdadinejad et al. [32] developed a model for coastal line extraction utilizing remote sensing data. Heidler et al. [2] introduced HED-Unet for the segmentation of coastal lines.

Therefore, obtaining coastlines using remote sensing imagery is feasible. Coastline segmentation algorithms can be categorized into traditional feature and deep feature methods. Their advantages and disadvantages are summarized in Table I, indicating that algorithms based on deep features have shown better performance and represent the primary direction of current research.

Recent challenges in coastal line segmentation using remote sensing images encompass the following: 1) the significant amount of remote sensing data complicates the process of annotating coastal lines; 2) the restricted capacity of single networks to capture and represent information; and 3) there is a limited amount of coastline data available for training purposes.

To address these challenges, a database of images and features is created utilizing Google Maps. The relationship between remote sensing imagery and annotated images serves as the foundation for establishing an image style mapping relationship through the utilization of attention-guided conditional generative adversarial network (AC-GAN) model. Dense Swin Transformer (DST) is utilized to establish the correlation between images and semantics. The integration of these methodologies enables the retrieval of coastal boundaries. Innovation points are given as follows:

1) utilizing an AC-GAN model that redefines the task of image segmentation by framing it as a style transformation problem;
2) constructing Dense Swin Transformer Unet (DSTUnet) model, which integrates Transformer to prioritize focal regions, thereby enhancing image and semantic interpretation;
3) proposing a transfer learning framework to integrate multiple features, leveraging the strengths of different networks to achieve accurate segmentation of coastal lines.

The study's structure is delineated as follows. Section II presents the algorithmic framework of this article. Section II-A provides a comprehensive explanation of AC-GAN model. Section II-B provides a comprehensive explanation of DSTUnet model. Section II-C introduces the fusion algorithm that relies on transfer learning. Section III delineates the data utilized in the experiments and the experiments conducted to validate the efficacy of the algorithms. Section IV provides a summary of the innovations and potential future research directions.

II. PROPOSED ALGORITHM

Remote sensing imagery enables the collection of data regarding coastal boundaries. The algorithmic workflow of this study is delineated in Fig. 1, primarily involving the following components.

1) Establishing a correlation between imagery and features to create a database.
2) This study introduces a coastal line extraction algorithm that utilizes an AC-GAN model. The algorithm establishes a style mapping relationship between input images and segmented images to facilitate coastal line extraction.
3) This study introduces a coastal line segmentation algorithm that utilizes DSTUnet. The algorithm aims to establish a connection between image and semantic representations to enhance coastal line extraction.

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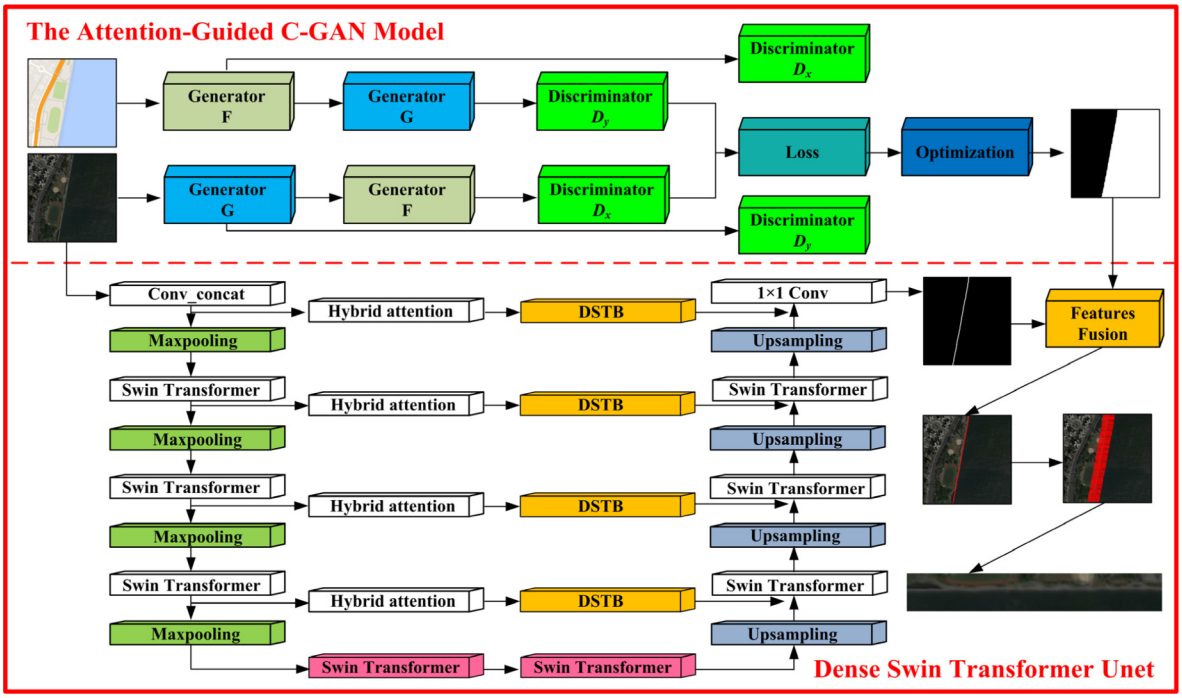


Fig. 1. Style transformation model for coastal line extraction using attention-enhanced CycleGAN.

4) A transfer learning algorithm is employed to combine the results from the previously mentioned algorithms to attain the ultimate coastal line extraction result.

A. Attention Guided C-GAN Model

Drawing inspiration from game theory, GAN architecture arises, characterized by its training methodology involving iterative and optimized processes akin to a game. The discriminator D assists the generator G in generating synthetic samples that align with authentic data distributions. Through optimizing both the discriminator and the generator, the generated samples closely resemble real samples, posing a challenge for the discriminator in distinguishing between the two distributions.

One of the most notable models is CycleGAN network, which is based on the principle of duality and involves the creation of a cyclical and inerrable structure. It consists of two mirror-symmetrical GANs [33], each containing two generators and two discriminators. This setup is designed to enable mutual style transformation between the source and target domains [34]. The core principle of CycleGAN network entails integrating a cyclic mechanism into the GAN framework, thereby liberating it from the requirement of paired training data. This innovation facilitates the transformation and constraint between two domains. Traditionally, achieving style transformation between two domains necessitated pairs of images from each domain with matching content to serve as training samples. To avoid the scenario where the transformer maps all images from one domain to a single image in another domain during feature transformation, a cycle consistency loss is utilized as a constraint. Building on the foundation laid by CycleGAN, researchers have undertaken comprehensive studies. Zhou et al. [35] utilized CycleGAN to reduce noise in low-dose images. Hammami et al. [36] proposed the development of a CycleGAN model for enhancing multiple organs in computed tomography (CT) images. Chen et al. [37] developed transformations for unsupervised video analysis. The superior performance of CycleGAN is evident, as the model necessitates a relatively smaller number of training samples.

This study reframes the issue of coastal line segmentation as a style transformation task. It integrates an attention mechanism into the CycleGAN network to create a style transformation model that can convert input imagery into annotated imagery for the purpose of coastal line extraction. The main workflow is outlined as follows.

1) *Binarization Process:* Utilizing Google Maps imagery to distinguish between marine and terrestrial regions, establishing a benchmark for coastal line delineation.
2) *Sample Selection and Processing:* From dataset X, select samples for processing by generator G, equipped with a dual-attention mechanism, to produce binary representations of marine and terrestrial zones in dataset Y, labeled as synthetic images $G(x)$.
3) *Image Domain Transformation:* Synthetic images $G(x)$ are reconverted to the original domain X using generator F, also featuring a dual-attention mechanism, resulting in $F[G(x)]$. These images are then assessed by their respective discriminators.
4) *Optimization Through Loss Evaluation:* Calculating loss to refine the performance of generators G and F and discriminators DX and DY for improved accuracy and efficiency.

The dataset was constructed by combining remote sensing imagery with binary annotated images, and C-GAN was employed for end-to-end training. The establishment of a

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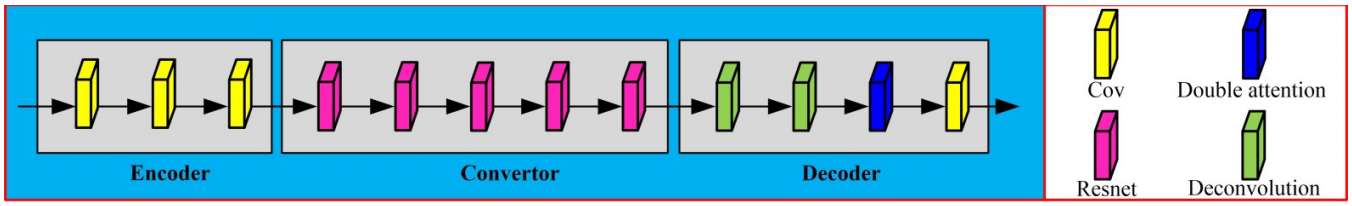


Fig. 2. D_Generator structure.

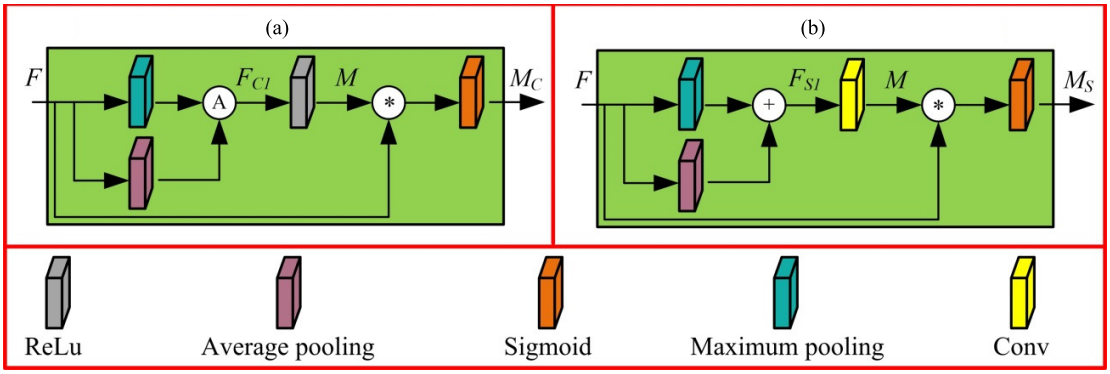


Fig. 3. Channel attention mechanism and spatial attention mechanism.

dual-attention generator, D_Generator, aims to convert and subsequently reconstruct the stylistic features present in the input images. Discriminators *DY* and *DX* are utilized to evaluate the extent to which the generated images adhere to the distribution of the source target images.

D_Generator comprises an encoder, a converter, and a decoder, as depicted in Fig. 2. The encoder utilizes three convolutional layers to extract diverse features from remote sensing images. Subsequently, the converter employs residual blocks for the processing of the feature vectors acquired in the preceding step. Finally, the decoder utilizes deconvolution, convolutional block attention module (CBAM), and convolutional layers to reconstruct the feature vectors into binary images.

The attention module comprises channel attention and spatial attention components. Channel attention is a mechanism that extracts features from the spectral dimension by assigning higher weights to salient features, thus focusing on them. Spatial attention is rooted in spatial structural information, focusing on salient features and assigning them greater weights.

The network architecture for channel attention [Fig. 3(a)] involves processing the input features F via average pooling and max pooling and subsequently transmitting them to the fully connected layer F_{C1}. After applying rectified linear unit (ReLU) operation, the resulting output M is obtained. The multiplication of matrix M with the feature vector F, followed by the application of the sigmoid function, results in the input features M_C for the spatial attention network.

The spatial attention model, illustrated in Fig. 3(b), receives F as its input. Subsequently, F undergoes global pooling and max pooling, followed by concatenation and dimensional reduction through a 7×7 convolutional layer. Activation via the sigmoid function leads to the occurrence of M_S. The final features are derived through the dot product of M_S and F.

The discriminator structure, as shown in Fig. 4, comprises multiple convolutional layers that extract features from images

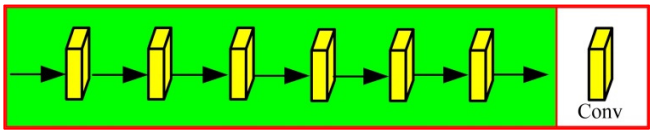


Fig. 4. Discriminator network structure.

produced by the generator. The process involves utilizing a 1-D output convolutional layer to assess the genuineness of the images, enabling the binary classification of real and generated samples.

The adversarial loss in C-GAN combines the losses from two GAN networks. This study utilizes the AC-GAN model, as depicted in Fig. 5. The loss function for generator G: X-> Y and its corresponding discriminator D_y is determined as follows:

$$L(G, D_Y, X, Y) = E_y\left[\log D_Y(y)\right] + E_x\left[\log(1-D_Y(G(x)))\right]. \tag{1}$$

The corresponding objective equation can be expressed as follows:

$$\min_G \max_{D_Y} L_{\text{GAN}}(G, D_Y, X, Y). \tag{2}$$

Given that generators G and F are identical, the objective functions for discriminator D_x, corresponding to generator F: Y-> X, can be represented as follows:

$$L(F, D_X, Y, X) = E_x\left[\log D_X(x)\right] + E_y\left[\log(1-D_X(G(y)))\right]. \tag{3}$$

The corresponding objective function is given as follows:

$$\min_F \max_{D_X} L_{\text{GAN}}(F, D_X, Y, X). \tag{4}$$

To preserve the structural integrity of the input image and retain the characteristics of the target image, a cycle consistency loss function is formulated. This guarantees that images x and y can be reconstructed back to the source domain

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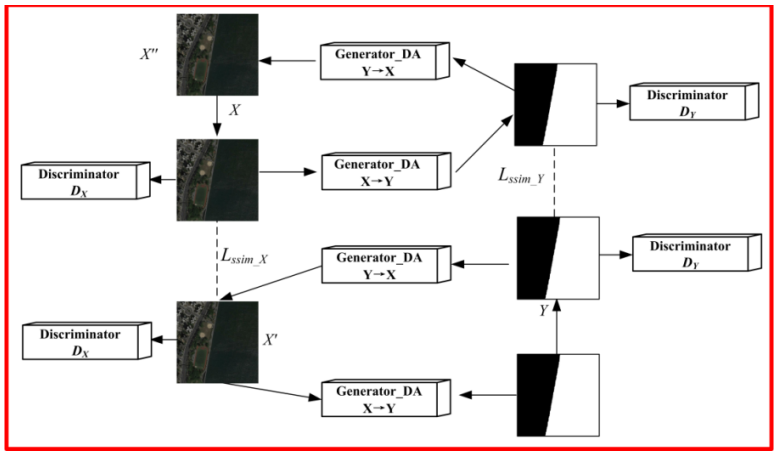


Fig. 5. AC-GAN network structure.

by the generator after being mapped from the source domain to the target domain. The cycle consistency loss function is expressed as

$$L_{\text{cyc}}(G, F, X, Y) = E_x\left[\|F(G(x) - x)\|_1\right] + E_y\left[\|G(F(y) - y)\|_1\right]. \quad (5)$$

The traditional C-GAN loss function is given as

$$L = L_{\text{GAN}}(G, D_Y, X, Y) + L_{\text{GAN}}(F, D_X, Y, X) + \lambda L_{\text{cyc}}(G, F, X, Y). \quad (6)$$

Given the certain similarity between the target and source domains, the structural similarity is shown as

$$\text{SSIM}(x, y) = l(x, y) * c(x, y) * s(x, y) \quad (7)$$

$$\begin{cases} l(x, y) = \dfrac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1} \\ c(x, y) = \dfrac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2} \\ s(x, y) = \dfrac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3} \end{cases} \quad (8)$$

where l represents the luminance; c represents the contrast; s represents the structure; u_x and u_y denote the means of x and y, respectively; and σ_{xy} represents the covariance of x and y. Given a constant c_n, the corresponding loss function is expressed as follows:

$$\begin{cases} L_{\text{ssim1}} = \left(1 - \text{SSIM}\left(X, X''\right)\right) + \left(1 - \text{SSIM}\left(Y, Y''\right)\right) \\ L_{1L_1} = \left\|X'' - X\right\|_1 + \left\|Y'' - Y\right\|_1 \\ L_{\text{ssim2}} = \left(1 - \text{SSIM}\left(X, X'\right)\right) + \left(1 - \text{SSIM}\left(Y, Y'\right)\right) \\ L_{2L_1} = \left\|X' - X\right\|_1 + \left\|Y' - Y\right\|_1. \end{cases} \quad (9)$$

The corresponding cycle consistency loss function is

$$L = \lambda_1 L_{\text{ssim1}} + \lambda_2 L_{\text{ssim2}} + \lambda_3 L_{1L_1} + \lambda_4 L_{1L_2}. \quad (10)$$

B. Dense Swin Transformer Unet

Unet utilizes a fully convolutional network for segmentation tasks. The symmetric structure of the network, which resembles the shape of a "U" and is named Unet [38], incorporates skip connections. These skip connections merge feature maps from the contracting path into the expansive path. End-to-end training is facilitated by Unet, which is a fully differentiable network. Owing to its architectural characteristics, Unet necessitates a relatively smaller amount of annotated data. This capability allows the network to extract image features while retaining detailed information, thereby achieving high precision in segmenting small objects. Extensive research has been carried out utilizing Unet. Zhu et al. [39] proposed a hybrid densely connected Unet architecture for segmenting medical lesions. Chen et al. [40] developed a channel-based Unet for object segmentation. Thomas et al. [41] introduced a multiple res-attention Unet model for object segmentation.

To enhance the algorithmic performance, researchers have undertaken studies focusing on the Transformer architecture. Originally utilized in the realm of language acquisition, Transformers have demonstrated significant accomplishments. Before the introduction of the Transformer model, the majority of neural network-based machine translation approaches depended on recurrent neural networks (RNNs), which employed recursion to perform sequential operations. Despite the commendable modeling capabilities exhibited by RNNs for sequences, their intrinsic sequential feature has led to slow training speeds and heightened training complexity, primarily attributable to their recurrent structure. In contrast, Transformers obviate the necessity of recursion by creating parallel structures to handle all data concurrently and utilizing self-attention mechanisms to fuse context with remote information. Transformers exhibit accelerated training rates and achieve optimal results compared to RNNs due to their ability to process information concurrently and holistically. Strudel et al. [42] harnessed the benefits of Transformers to create a connection between images and semantics. Gao et al. [43] developed a Transformer model for image segmentation. Chen et al. [44] proposed a multilevel attention-guided Unet with Transformer for image segmentation.

The study integrates Unet and Transformer to create a network architecture that includes encoders, decoders, and skip connection layers. The skip connection layers are comprised of dense ST blocks (DSTBs) and hybrid attention modules, which serve to improve the exchange of information between encoders and decoders.

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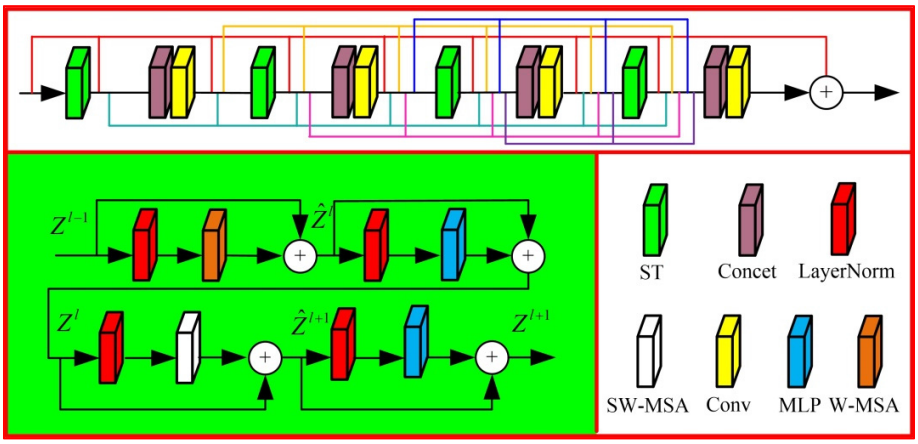


Fig. 6. DST structure.

Contemporary Unet models frequently utilize simple skip connections, which possess restricted scope and can result in notable semantic disparities and loss of information. To address this issue, DSTBs and hybrid attention modules are incorporated into the skip connection pathways to amplify the model's long-range dependencies and facilitate the efficient transfer of feature information between encoders and decoders.

Drawing inspiration from dense connectivity, the skip connection pathways incorporate DSTBs, where each DSTB consists of multiple Swin Transformer (ST) blocks (Fig. 6).

Each Transformer transmits feature information to the subsequent ST by utilizing shifting windows to enable multihead self-attention modules. Within two consecutive Transformer modules, window-based multihead self-attention (W-MSA) modules and shifted W-MSA (SW-MSA) modules, are employed. The expressions for these are expressed as

$$\begin{cases} \hat{z}^l = \text{WMSA}\left(\text{LN}\left(z^{l-1}\right)\right) + z^{l-1} \\ z^l = \text{MLP}\left(\text{LN}\left(\hat{z}^l\right)\right) + \hat{z}^l \\ \hat{z}^{l+1} = \text{SWMSA}\left(\text{LN}\left(\hat{z}^l\right)\right) + \hat{z}^l \\ z^{l+1} = \text{MLP}\left(\text{LN}\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1}. \end{cases} \tag{11}$$

The window attention is defined as

$$\text{Attention}(Q, K, V) = \text{soft max}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \tag{12}$$

where Q, K, and V represent matrices; d denotes the dimensionality of Q and K; and B signifies the bias. The output of each ST is

$$F_i^t = P_{\text{ST}}\left(P_{\text{FF}}\left(F_i, F_i^1, \ldots, F_i^{t-1}\right)\right) \tag{13}$$

where P_{ST} and P_{FF} represent ST operations and feature fusion operations, respectively.

C. Transfer Learning Algorithm

Upon receiving the results generated by the AC-GAN model and DSTUnet model, it is essential to merge these results. Common fusion strategies involve pixel-level fusion, where the pixels of the entire image are merged directly. The size of image data is directly correlated with the processing time of the algorithm. Unprocessed data accumulate both valuable information and noise, which can diminish the quality of the final result. Pixels, which are prone to noise, produce unreliable results. Feature-level fusion involves the integration of distinctive features such as edges, shapes, and contours to establish a comprehensive representation, frequently incorporating specific prior knowledge. Decision-level fusion involves combining the outputs of multiple sensors after each sensor has made its independent decision, thereby improving the overall decision-making process.

The binary image obtained from the AC-GAN model is C_{img}, and the coastal line probability image from DSTUnet is S_{img}. Employing a feature-level fusion strategy with parameter λ, the coastal line probability is determined as follows:

$$U_{\text{img}}(x, y) = \lambda C_{\text{img}}(x, y) + (1 - \lambda) S_{\text{img}}(x, y). \tag{14}$$

A threshold T is set to label the satisfying results. Feature-level fusion may result in discrete points. Given the continuity of coastal lines, an initial step involves the application of mathematical morphological operations, which can be specifically expressed as follows:

$$C = (A \oplus B) \odot B \tag{15}$$

where A represents the binary image, B represents the structuring element, $\oplus$ represents the dilation, and $\odot$ represents the erosion.

Adjacent points are linked through common connectivity region labeling algorithms such as four-connectivity and eight-connectivity (Fig. 7). In this representation, the central point is denoted by black color, while the connected region is depicted in lighter shades.

Regions characterized by a greater number of interconnected pixels are preserved during the process of coastal line extraction. Given the constraints of the limited database utilized in this research and the necessity to apply the algorithm to data gathered from different detectors for generalization purposes, the utilization of transfer learning is implemented to attain the ultimate coastal line extraction results. The framework proposed in this article mainly consists of attention C-GAN and Swin-Unet models. Our strategies include two approaches: 1) one of the models undergoes transfer learning, where the learned parameters are fixed, and the parameters of the other model are dynamically adjusted; and 2) transfer learning is applied separately to train both models, obtaining parameters that are adjusted for each model. The data used in this study

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TABLE II
RESULTS OF AC-GAN DIFFERENT CONNECTION METHODS

Data	Method	Type	*AOM*↑	*AVM*↓	*AUM*↓	*CM*↑
Aerial photo-Maps	Serial	1. Spatial 2. Channel	0.78	0.15	0.12	0.84
		1. Channel 2. Spatial	0.71	0.21	0.18	0.77
	Parallel	λ=0.3; λ=0.7	0.75	0.16	0.15	0.81
		λ=0.5; λ=0.5	0.82	0.11	0.14	0.86
		λ=0.7; λ=0.3	0.87	0.09	0.10	0.89
Ziyuan 3	Serial	1. Spatial 2. Channel	0.73	0.18	0.17	0.79
		1. Channel 2. Spatial	0.66	0.23	0.20	0.74
	Parallel	λ=0.3; λ=0.7	0.71	0.19	0.22	0.77
		λ=0.5; λ=0.5	0.76	0.16	0.12	0.83
		λ=0.7; λ=0.3	0.80	0.13	0.11	0.85

are sourced from the dataset provided in the earlier stages of this research.

III. EXPERIMENT ANALYSIS

The data chosen for this study consist of two primary components, as illustrated in Fig. 8. The initial section comprises 320 pairs of images depicting coastal lines alongside their annotated counterparts sourced from the Aerial photo-Maps dataset, with an image resolution of 600 × 600. The second part comprises RGB images obtained in Sri Lanka from Ziyuan 3, with a resolution of 885 × 8976, totaling 13 frames. The experimental model employs the TensorFlow framework on a Linux operating system and is trained using a GTX2060 16 GB GPU. The ratio of training data to testing data is kept at 1:1.

A. Parameter Selection

Experiments were conducted to obtain optimal parameter results, with a focus on: 1) the connection method of the attention mechanism and 2) the approach to transfer learning.

The performance of the algorithm is evaluated using the area overlap measure (AOM), area oversegmentation measure (AVM), area undersegmentation measure (AUM), and combination measure (CM)

$$\begin{cases} \text{AOM} = \dfrac{P \cap Q}{P \cup Q} \\ \text{AVM} = \dfrac{P - Q}{P} \\ \text{AUM} = \dfrac{Q - P}{Q} \\ \text{CM} = \dfrac{1}{3}\{\text{AOM} + (1 - \text{AVM}) + (1 - \text{AUM})\} \end{cases} \tag{16}$$

where P represents the gold standard, Q denotes the algorithmic segmentation results, AOM and CM are directly proportional to algorithm performance, and AVM and AUM are inversely proportional.

In the algorithm for extracting coastal lines based on AC-GAN, a comparison of the impacts of serial and parallel algorithms is presented in Table II. The performance achieved

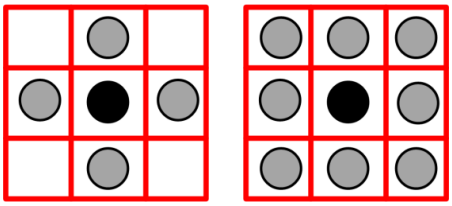


Fig. 7. Connectivity region labeling with four-connectivity and eight-connectivity illustration.

on Google dataset exceeds that of Sri Lanka dataset. This disparity emerges due to the algorithm being trained on Google data, which results in a bias toward the features present in the Google dataset. The approach that sequentially applies channel attention followed by spatial attention in serial arrangements demonstrates the lowest level of effectiveness compared to parallel arrangements. This result can be attributed to the channel attention mechanism reducing the spatial correlation among features, which consequently results in decreased performance in the regions emphasized by spatial attention. The strategy of implementing spatial attention prior to channel attention is the second most effective approach, as spatial attention diminishes the importance of channel attention in a similar manner. However, the stronger correlation of spatial attention leads to superior performance in comparison to the scenario where channel attention precedes spatial attention.

In the parallel approach, spatial and channel attentions are computed independently and subsequently integrated. Various fusion weights are being compared. When the weight of spatial attention is set to $\lambda = 0.3$ and the weight of channel attention is set to $\lambda = 0.7$, the performance decreases. As the proportion of spatial attention increases, with both spatial and channel attentions weighted equally at 0.5, performance improves, suggesting the equal significance of both types of attention. The optimal results are attained when assigning a weight of 0.7 to spatial attention ($\lambda = 0.7$) and 0.3 to channel attention ($\lambda = 0.3$), highlighting the predominant influence of spatial attention over channel attention in the result.

Table III displays the results of the feature fusion algorithm, with the check mark indicating the fine-tuning algorithm. Employing a parallel approach with a spatial attention weight

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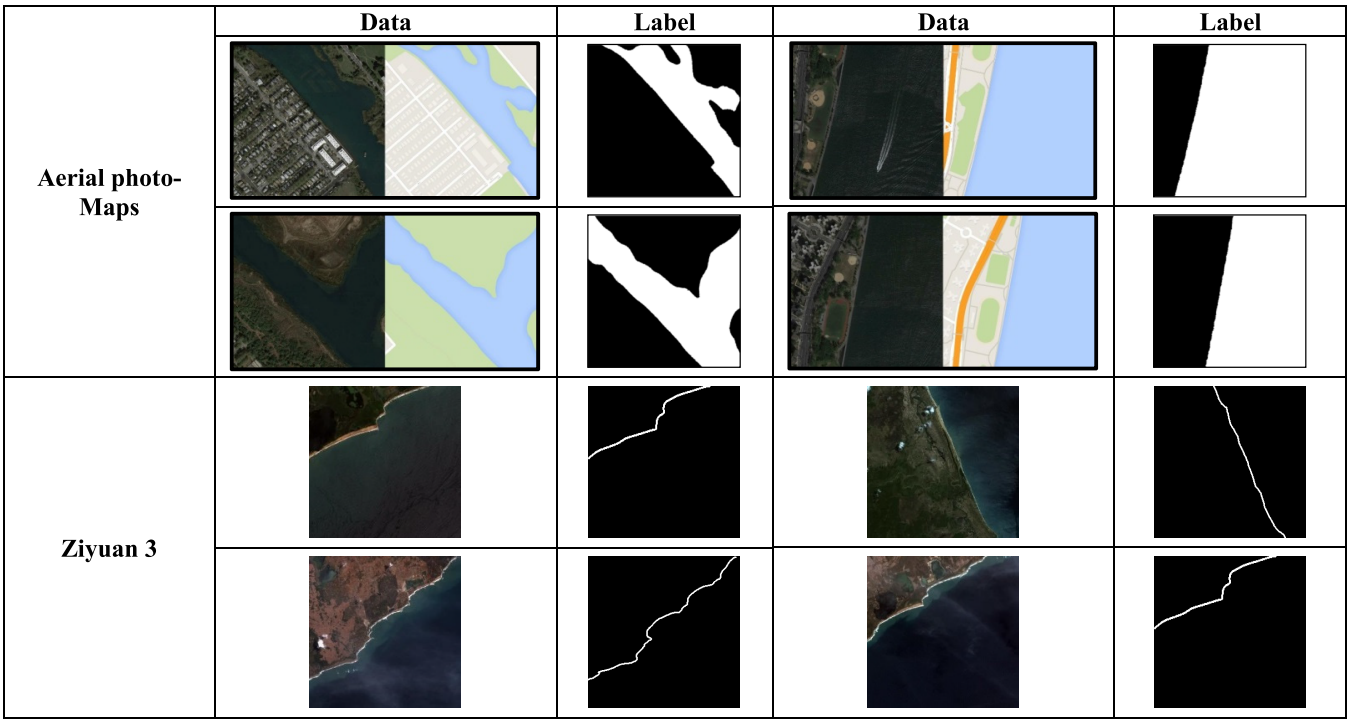


Fig. 8. Two primary components of selected database.

TABLE III
RESULTS OF FEATURE FUSION ALGORITHM

Data	Fine-tuning		*AOM*↑	*AVM*↓	*AUM*↓	*CM*↑
	AC-GAN	DSTUnet				
Aerial photo-Maps	√	√	0.90	0.08	0.10	0.90
	√		0.88	0.12	0.09	0.89
		√	0.92	0.06	0.07	0.93
Ziyuan 3	√	√	0.83	0.12	0.13	0.86
	√		0.81	0.15	0.11	0.85
		√	0.85	0.10	0.08	0.89

$\lambda = 0.7$ and a channel attention weight $\lambda = 0.3$, the effectiveness of transfer learning is validated. When both AC-GAN and DSTUnet models undergo fine-tuning, the pre-trained features of AC-GAN are underutilized, leading to only marginal enhancements. When exclusively fine-tuning AC-GAN, coastal zone segmentation can be accomplished by keeping DSTUnet training features fixed. When repairing AC-GAN and fine-tuning DSTUnet, the features of AC-GAN training are maximally leveraged, resulting in optimal performance.

Fine-tuning both AC-GAN and DSTUnet based on training with Aerial photo-Maps dataset results in decreased performance. This result can be attributed to the Aerial photo-Maps dataset being initially trained exclusively on AC-GAN, enabling the acquisition of relevant features. However, DSTUnet was inexperienced initially. The incorporation of novel data can disrupt existing knowledge, resulting in decreased performance efficiency. Fine-tuning DSTUnet while keeping AC-GAN fixed leads to improved results through the continued exploration of features derived from learned characteristics, as depicted by the convergence curve in Fig. 9.

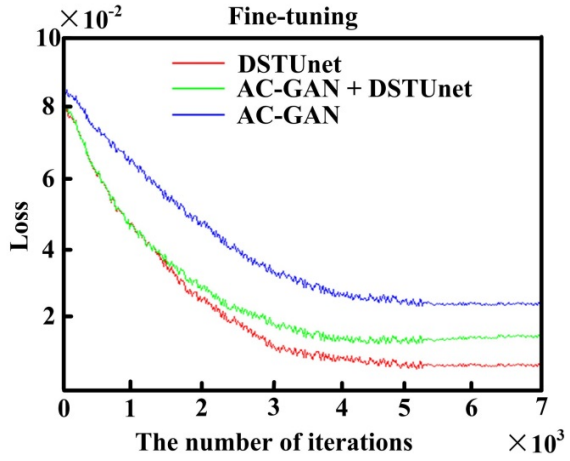


Fig. 9. Evolution of algorithm performance across iterations.

B. Comparative Results of Algorithms

This study compares a conventional feature algorithm as thresholding [19] and a boundary-based method [26] with deep learning algorithms as Unet [38] and HED-Unet [2]. The algorithm described in the literature [19] utilizes manual

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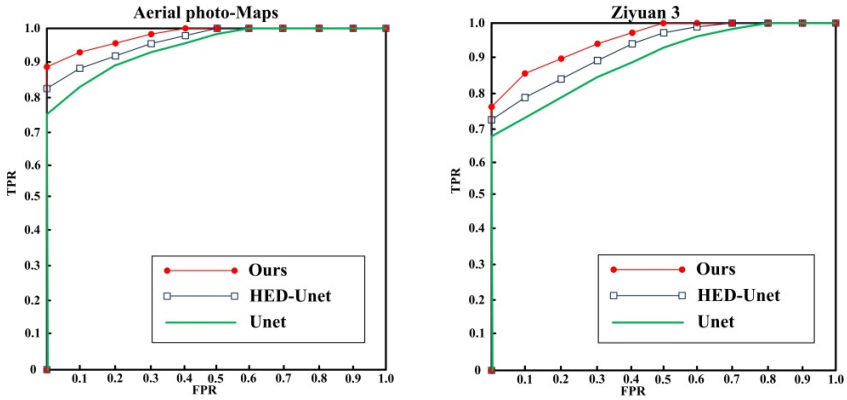


Fig. 10. ROC results of different algorithms.

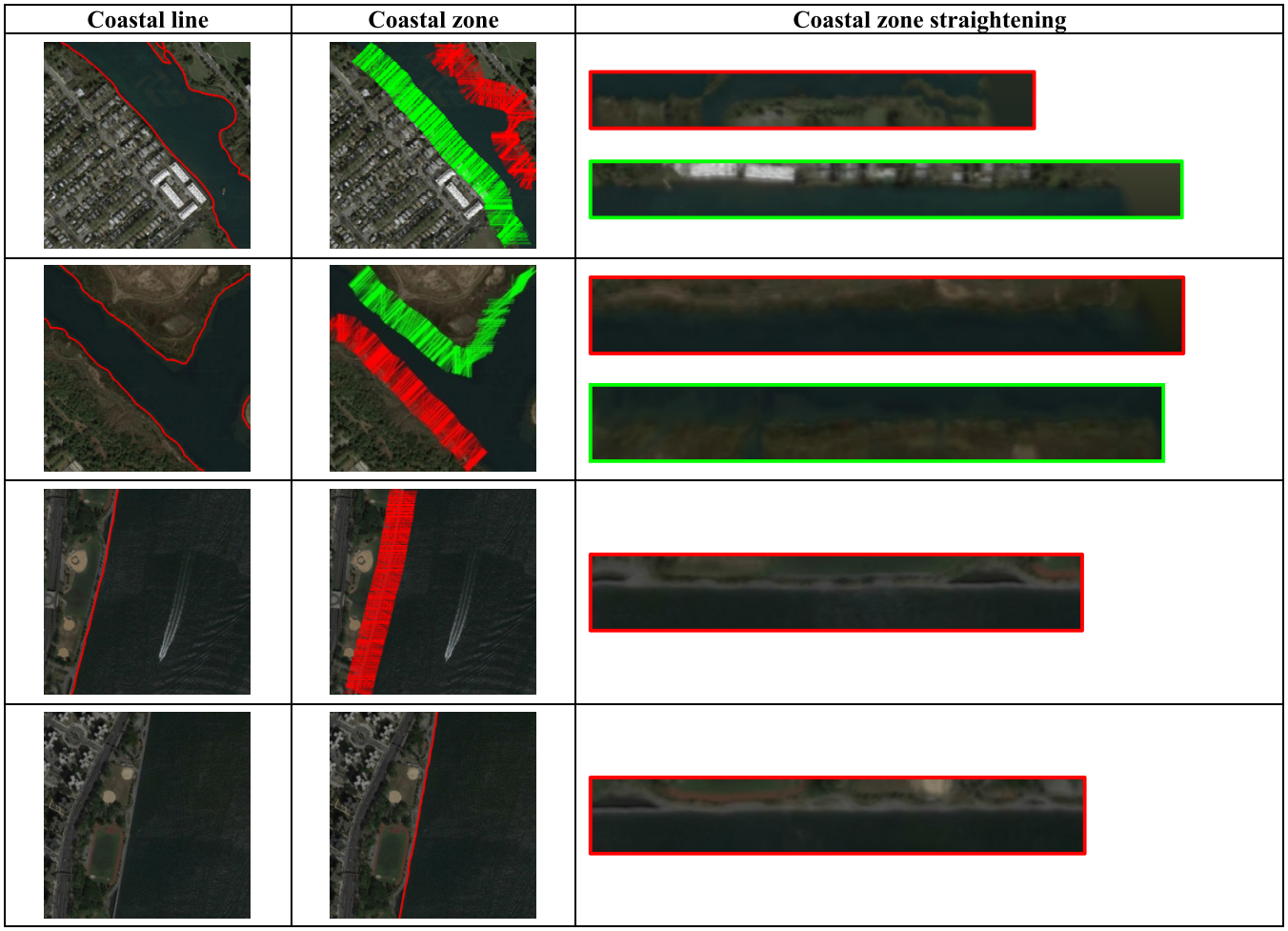


Fig. 11. Results of coastal line segment by Aerial photo-Maps.

interaction for the selection of precise color thresholds for segmentation. Modava and Akbarizadeh [26] introduced a boundary model based on an energy perspective to achieve segmentation of coastal lines. However, these two algorithms require high-quality data and provide fast computation speeds, yet they do not possess the capability to process complex, large-scale imagery. Huang et al. [38] employed a Unet-based model to establish a correlation between imagery and semantics, demonstrating a certain level of effectiveness. Heidler et al. [2] incorporated segmentation and edge detection into Unet via HED-Unet, thereby enhancing the network's attention and subsequently enhancing the algorithmic performance. The algorithm presented in this article formulates models focusing on style transformation and semantic description, integrating an attention mechanism to attain optimal results. Comparisons from Table IV suggest that Aerial photo-Maps outperforms Ziyuan 3. This can be attributed to the algorithm being trained on a subset of Aerial photo-Maps data and the relative simplicity of Aerial photo-Maps data, which enhances the algorithm's applicability. Ziyuan 3 covers

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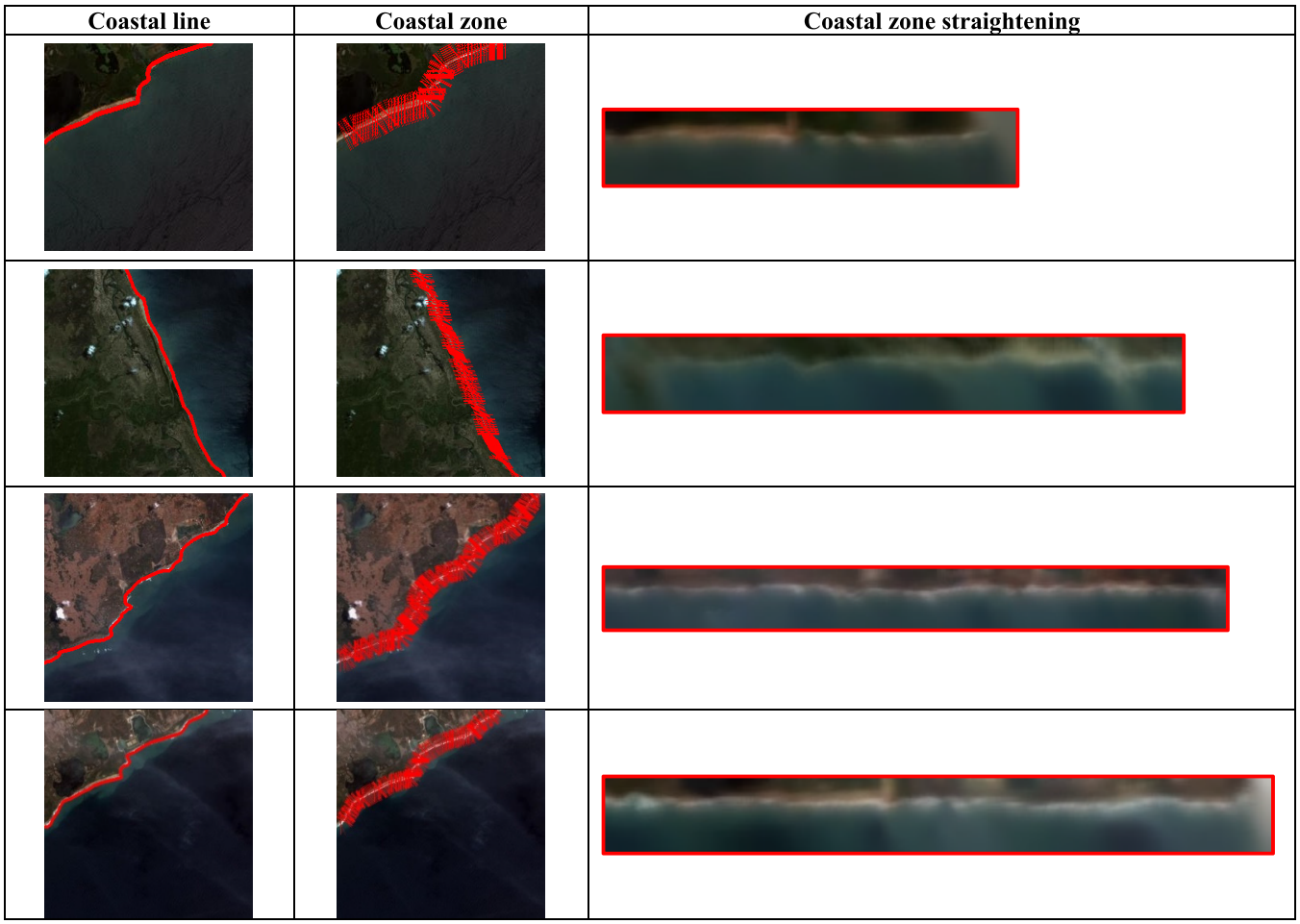


Fig. 12. Results of coastal line segment by Ziyuan 3.

TABLE IV
RESULTS OF DIFFERENT ALGORITHMS

Data	Algorithm	*AOM*↑	*AVM*↓	*AUM*↓	*CM*↑
Aerial photo-Maps	Thresholding [19]	0.73	0.17	0.19	0.79
	Boundary [26]	0.78	0.14	0.15	0.83
	Unet [38]	0.85	0.13	0.14	0.86
	HED-Unet [2]	0.88	0.10	0.09	0.90
	Ours	0.92	0.06	0.07	0.93
Ziyuan 3	Thresholding [19]	0.73	0.19	0.18	0.78
	Boundary [26]	0.77	0.16	0.17	0.81
	Unet [38]	0.80	0.15	0.14	0.84
	HED-Unet [2]	0.82	0.13	0.12	0.86
	Ours	0.85	0.10	0.08	0.89

a broader range of perspectives, incorporating a variety of features within a multifaceted environment. In brief, although all algorithms can accomplish coastal line segmentation, conventional algorithms exhibit restricted representational capacities and struggle to significantly improve performance. Deep learning-based algorithms exhibit enhanced representational capacities, leading to superior results. The study presents receiver operating characteristic (ROC) curves comparing deep learning-based algorithms with the algorithm proposed in this article, as illustrated in Fig. 10. The Unet algorithm demonstrates a certain level of effectiveness, while the HED-Unet algorithm displays enhanced performance. The proposed algorithm optimizes the benefits of multiple models, yielding the optimal results.

C. Algorithm Performance Demonstration

For the intuitive display of algorithm performance, we centered on the coastline extracted by the algorithm, took 50 pixels as the radius, and unfolded the coastline using linear interpolation. The Aerial photo-Maps data contain multiple coastline images with significant curvature changes, as shown in Fig. 11. The Ziyuan 3 data contain longer coastlines as

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shown in typical results in Fig. 12. It can be observed from the figure that the proposed algorithm can effectively segment coastlines, with the sea and land concentrated on the two sides of the image, validating the algorithm's effectiveness.

During the experimental process, it was found that straightened images appeared blurry, which was due to the linear interpolation algorithm used. In the future, we will adopt a deep learning approach to achieve super-resolution of images for high-definition display of coastlines.

IV. Conclusion

The study focuses on the importance of coastal line segmentation by employing remote sensing techniques to obtain imagery and to carry out research on coastal zone segmentation, resulting in specific achievements. The innovations comprise the following.

1) The problem of image segmentation is redefined as image style transformation by proposing a coastal line extraction algorithm that utilizes the AC-GAN model.
2) The integration of Transformer technology into Unet for the development of a dense framework has led to the proposal of a coastal line segmentation algorithm based on DSTUnet. This advancement aims to enhance image processing and semantic description capabilities.
3) In light of the constraints posed by limited samples, this study proposes the implementation of a transfer learning framework and the establishment of a multifeature fusion mechanism. These strategies aim to maximize the advantages of the network and ultimately achieve accurate coastal line segmentation.

However, the segmentation process continues to face persistent challenges, necessitating further research. One such challenge is the difficulty in acquiring and annotating remote sensing imagery, which leads to limited data availability. In addition, the variety of annotated coastal line styles is constrained, highlighting the need for enhancements in the algorithm's capacity for generalization. Future research endeavors will investigate these facets.

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